

Wang Zhaoqiang

Job Objective: Algorithm Engineer

Intended city: Xi'an



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Education experience

2012.09 – 2015.04	Northwest University	master	Computer technology
2008.09 – 2012.06	Northwest University	bachelor	Information and computing science

Work experience

2015.04– 2026.01	Xi'an Huawei Technology Co., Ltd.	Algorithm Engineer	Search advertising NLU, computer vision, network planning and optimization
2026.01– Present	LONGi Green Energy Technology Co., Ltd.	Algorithm Engineer	Photovoltaic (PV) Production Process Parameter Optimization; Agent Development

Professional skills

- ◆ **Programming language framework:** Proficient in programming languages such as Java and Python, familiar with TensorFlow, PyTorch, and ONNX. Proficient in the Runtime framework and skilled in using libraries such as cv2, pillow, and pandas.
- ◆ **Database:** Familiar with non-relational databases such as DCS and OBS, as well as vector databases like Elasticsearch and Faiss.
- ◆ **NLP algorithms:** BERT, BGE, Transformer architecture, model pruning, LoRA fine-tuning, familiar with the application and implementation of large model-related technologies (LLM, RAG, Agent)
- ◆ **Computer Vision:** Familiar with video/image quality assessment (TRIQ, rankIQA), object recognition, feature extraction (VGG, RESNET), etc.
- ◆ **Network planning and optimization:** Familiar with search algorithms such as GA, ant colony algorithm, and simulated annealing, matching algorithms like HMM and DTW, machine learning algorithms, and proficient in modeling languages such as R and MATLAB.

project experience

Crystal Pulling Breakage Rate Data Q&A Agent 2026.06 – Present

Project Background: In monocrystalline silicon production, wire breakage during crystal pulling is a key yield-killing anomaly — it leads to ingot scrapping and thermal field damage, with losses ranging from thousands to tens of thousands of RMB per furnace run. I built a self-developed Q&A analysis Agent based on LangGraph, enabling engineers to perform self-service breakage-rate analysis across multiple production bases, furnaces and product models in natural language, with permission-filtered results returned in seconds.

Tech Stack: Python, LangGraph, FastAPI, BM25/Vector Retrieval, Function Calling (MCP)

Key Work:

1. LangGraph workflow orchestration: Designed a multi-node directed graph of "Preprocessing Routing → Schema Retrieval → SQL Agent Reasoning & Execution → Result Generation," passing context across nodes through a TypedDict State; implemented human-in-the-loop collaboration via LangGraph interrupt, allowing the graph to resume from checkpoints after user confirmation without context loss.
2. Three-level Schema Linking retrieval: Built dual-path indexes (BM25 sparse retrieval + dense vector retrieval) over a 5-table database schema, fused and ranked with RRF, and recalled relevant tables and fields in a structured manner by metrics / dimensions / filter conditions — solving the problems of incorrect table/field selection and missed JOIN relationships in Text2SQL over large schemas.

3. ReAct tool-calling loop (MCP protocol): Encapsulated SQL execution, current-time lookup and date-expression parsing as standard MCP tools registered on a tool whitelist; through Function Calling, the Agent autonomously drives a multi-round reasoning loop of "parse time → generate SQL → execute → read error → fix and retry," with a maximum-round cap as a fallback.

Business Impact:

1. Covers 8+ types of high-frequency analytical questions (e.g., 30-day daily breakage-rate trends), returning results and analytical conclusions end-to-end in seconds from natural-language questions.
2. Permission isolation across three roles: unauthorized tables are unreachable at the retrieval, generation and execution layers.
3. Fully decoupled the Agent framework from the business schema — table structures, role permissions and SQL generation rules are all configuration-driven; the framework has been validated for rapid reuse by switching to a different industry domain.

PECVD & Screen-Printing Offset Parameter Optimization 2026.01 – 2026.06

Project Background: Parameter optimization for the coating and screen-printing processes of solar cells.

Addressing two issues — large film-thickness variation in PECVD coating, and broken/missing grid lines caused by screen-printer head offset from mechanical vibration — the project achieved automatic parameter optimization and closed-loop calibration through a combination of DOE process experiments and a software system.

Tech Stack: Flask / Java / Python / YOLOv8 / MySQL / FTP / Multithreading

Key Work:

1. PECVD temperature-zone optimization and scheduling service: Designed DOE experiment plans, systematically tuning the temperature curve of each zone and the SiH_4 gas-flow ratio to keep front-side coating thickness stable at 65 ± 5 nm; developed a scheduling service that centrally manages process-parameter distribution and data collection across multiple bases and machines.
2. Intelligent screen-offset detection and calibration: Used YOLOv8 to detect five alignment points (A/B/C/D/center) and quantify printing deviation, and designed a geometric adjustment algorithm to calculate calibration offsets (X/Y/rotation angle); integrated SCADA for equipment-status reading and MQTT for calibration-command dispatch with asynchronous confirmation, forming a closed loop of detection → decision → dispatch → verification.
3. High-availability architecture: Built a multithreaded task-coordination framework managing five periodic services including FTP image acquisition, batch inference and parameter distribution; implemented a three-level FTP connection pool (idle reclamation + dynamic refresh), MQTT automatic reconnection, and a database connection pool with failure retries to ensure stable operation; four core tables track full-pipeline data, with DingTalk alerting and YAML-driven multi-base deployment.

Business Impact:

1. Film-thickness uniformity CV reduced from 4.2% to 1.8%; broken/missing-grid defect rate reduced from 2.3‰ to 0.6‰; cell efficiency (ETA) improved by 0.08%.
2. Offset calibration upgraded from manual inspection (~15 min) to fully automatic real-time calibration (within 30 seconds).
3. Deployed across 2 bases and 50+ machines, delivering annual cost savings of approximately RMB 1.5 million.

Search Advertising Relevance Admission Microservice 2025.01 - 2026.01

Project Background: To build a large-scale semantic matching system with millisecond-level response, it is necessary to simultaneously meet the following requirements:

- ◆ High accuracy: Screen out more than 90% of relevant advertising materials from over 4 million advertising materials.
- ◆ Low latency: Over 400 million inference requests per day, with end-to-end P99 latency < 30ms

Technology stack: Java / Python / BERT / DCS / OBS / LLM

Core technology breakthroughs:

1. A large model-driven query semantic understanding system, pioneering a dual-model architecture of "BERT + Qwen":
 - ◆ Online microservice layer: The Sentence-BERT model, after applying structured pruning (retaining 3 layers) and dynamic quantization, achieves semantic vector extraction within 6ms.

- ◆ Offline periodic task layer: Deploy Qwen2-14B through LangChain and enhance the advertising domain capabilities with RAG. Mainly involves the segmentation of APP materials. FAISS database setup and optimization; RAG supports multi-path recall (vector and keyword), and simultaneously uses 3% of the traffic data for automatic reflow to train BERT. The first version material identification F1 has improved by 3.2%.

2. Engineering performance optimization

- ◆ NPU acceleration adaptation: Convert ONNX models to OM format and utilize 910b to achieve NPU inference (5.6 -> 1.2 ms) with ultra-low latency.
- ◆ LRU cache architecture: Build a two-level cache strategy of Caffeine (local memory) → DCS (distributed memory), reducing the access pressure on DCS by 96%.

Main achievements:

- ◆ The F1 accuracy of RAG-related material identification has increased by 3.2%;
- ◆ Within half a year, we have increased the ECPM of advertisements by 13% and raised the daily revenue by 230,000 yuan (from 1.72 million yuan to 1.95 million yuan).

Search Advertising NLU Microservice Optimization 2023.5 - 2024.12

Project Background: Faced with a business bottleneck of an average daily transaction volume of 500,000, led the transformation of a Python microservice with a CPU utilization rate of only 5%.

Performance optimization: The word segmentation weight module uses the traditional BERT model, which has a large number of parameters (110M), resulting in an online inference latency as high as 42ms; the understanding of long-tail queries relies on rule supplementation, with high maintenance costs.

Semantic understanding: The F1 score for intent recognition in semantic understanding is less than 85%, which affects the recall accuracy and the conversion rate of advertisements.

Architecture adjustment: A processing capacity gap of 25 million queries per month

Technology stack: Java/Python/BERT/DCS/OBS/LLM

Core technology breakthroughs:

1. Offline computing of large models

- ◆ Innovative Dispatcher-Worker Distributed Dynamic Architecture: The resource scheduling system achieves CPU utilization of 98% and GPU utilization of 83%(originally 30%), completing full processing in 48 hours with 5 nodes (originally requiring 30 days); it covers 86% of high-frequency query requests, reducing online load by 60%.
- ◆ Qwen large model SFT: By meticulously constructing 5,000 hard cases of NER through deepseek-r1 and fine-tuning Qwen2-14B with LoRA, the F1 score was improved by 17% at the SPU level compared to the online model.
- ◆ Designing structured prompt engineering: Through goal customization, few-shot positive and negative examples, metric calculation, and format control, ensure the application and implementation of large models.

2. High-performance online services

- ◆ Adopt a Java microservice layered design (interface gateway/business logic/cache acceleration), and implement dynamic data update using the listener pattern.
- ◆ Lightweight BERT: Through structured pruning and dynamic quantization, the parameter size is compressed to 5% of the original, and the latency is reduced to 6ms.
- ◆ Simultaneously, it is adapted to replace the BERT embedding model with bge-small, increasing the material recall rate by 23%.

Main achievements:

- ◆ Algorithm metrics: Intent recognition F1 score 91.5% (up by 6.5%), offline SPU entity recognition F1 score increased by 17%.
- ◆ Performance indicators: P99 latency reduced by 26% (from 42ms to 31ms), resource cost decreased by 50% (saving 1200 cores)
- ◆ Business value: Within one and a half years, the daily transaction volume was driven from 500,000 to 1.72 million (a 244% increase).

Huawei Video UGC Content Deduplication System 2020.4 - 2021.8

Project Background: To address the issue of duplicate push of 11 million UGC videos on Huawei's video platform, which includes various scenarios such as duplicate original videos and mixed-cut splicing, the team has developed a multimodal deduplication system with millisecond-level response. The system is required to meet the technical requirements of a time difference of less than 1 second for duplicate fragment location and an overall deduplication accuracy rate of over 95%.

Technology stack: Python / Video fingerprint / Audio fingerprint / faiss / CV

Core technical solution:

1. Multi modal fingerprint system

- ◆ Visual processing: A key frame feature extraction pipeline is built based on the VGG16 architecture, and the Faiss Poisson engine is combined to achieve efficient retrieval of billions of vectors (the IVF index construction time is controlled within 10 minutes).
- ◆ Audio processing: Improved the Vox audio fingerprint algorithm, optimized MFCC feature extraction for scenarios with voice alteration and background music interference, achieving a recall rate of 97% in four types of voice alteration test sets.
- ◆ Temporal matching: We innovatively developed the Clip-DTW sequence matching model, effectively solving the problem of segment localization in mixed-cut videos. Its error accuracy has reached the industry-leading level (verified by comparison with Alibaba under the same standards).

2. Service architecture design

- ◆ Adopt a three-level filtering mechanism: coarse screening layer (Faiss nearest neighbor search) → fine screening layer (time series correlation analysis) → positioning layer (DTW dynamic time warping).
- ◆ Build a microservice system based on the Flask framework, stably processing 800,000 requests daily.

Main achievements:

- ◆ Full test set validation metrics: Precision 96.6%, Recall 97.08% (outperforming the competing solutions from CloudBU and Peking University's School of Electronics Engineering and Computer Science)
- ◆ Extended technical application: Successfully migrated to the duplicate checking and video review scenarios of Huawei Music's MV library, reducing manual review costs by 30%.

Research on Key Technologies for Intelligent Parameter Optimization of Wireless Networks 2015-2020

Project Background: Wireless Network Characteristic Parameter optimization A global optimization problem, by adjusting the parameters of adjacent cells to avoid interference, balance the load, and ultimately improve user throughput. Features involved: Interference Randomization Optimization (IR), Mobile Load Balancing (MLB), ICIC. General network parameter tuning involves a large number of cells and cannot be solved in polynomial time. Scenario modeling is required to construct an evaluation function, and then a heuristic search method is used to approximate the optimal solution. Virtual road tests are designed to address the high cost of road test data, and MR+ location prediction is used as an alternative. Intelligent site planning is to identify the cell to be planned and determine the range of RF parameters for optimization.

Technology stack: Python/Scala/R/Matlab/ Heuristic Search (GA)/Sequence Prediction HMM/ Clustering/Machine Learning

Core technological innovations:

1. Interference Randomization Optimization (IR)

- ◆ Developed an optimization scheme for RB resource allocation based on genetic algorithm, achieving a 4.5% increase in throughput at 1000 stations through co-frequency small segmentation clustering (DBSCAN improvement) and interference matrix modeling
- ◆ Innovation: Design equal-length gene coding (2 bits/cell) and weighted evaluation function to solve the 4^n solution space search problem

2. Mobile Load Balancing (MLB)

- ◆ Build nonlinear load-rate models and develop automated parameter optimization systems to replace manual experience
- ◆ Achieved an 8.88% increase in road test throughput, with the solution implemented in over 30 cities

3. Virtual road test locations

- ◆ MR Localization algorithm based on Hidden Markov Model (HMM) through road rasterization modeling and DTW path calibration
- ◆ Achieving 84% accuracy and 91% recall, reducing 80 percent of field road test work

Main achievements:

- ◆ Scale application: All solutions complete live network validation and GENEXCloud plug-in
- ◆ Efficiency improvement: Parameter optimization time reduced from 3 days manually to 10 minutes
- ◆ Quality standards: The key point is that the standard call rate exceeds 90% of the acceptance line